

Cryptocurrency Movement Prediction Using Machine Learning and Time-Series Models

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Abstract—Cryptocurrency markets are highly volatile, making accurate forecasting a persistent challenge. Traditional financial models often fall short in capturing nonlinear patterns in price movements, while modern machine learning approaches have shown increasing promise. In this study, I conduct a comprehensive comparative analysis of three distinct forecasting models to predict the directional movement of Bitcoin and Ethereum: Logistic Regression, ARIMA, and Long Short-Term Memory (LSTM) neural networks. Unlike prior research that typically focuses on a single model or emphasizes exact price prediction, this work evaluates directional classification accuracy to provide meaningful insights for real-world trading contexts. Using five years of daily OHLCV data, I implement each model, analyze performance through accuracy, precision, recall, and F1-score, and visualize confusion matrices and ROC curves. The results demonstrate that no single model universally dominates; however, LSTM shows superior recall for upward movements, while ARIMA provides more balanced directional predictions. This study contributes a practical evaluation of traditional, statistical, and deep learning approaches within the same experimental framework and highlights the strengths and limitations of each for crypto trend forecasting.

Index Terms—Cryptocurrency, Machine Learning, ARIMA, LSTM, Bitcoin, Ethereum, Time-Series Forecasting, Trend Prediction

I. INTRODUCTION

Cryptocurrency has evolved from a niche technological experiment into a major global financial asset class, with a market capitalization exceeding four trillion dollars. As cryptocurrencies such as Bitcoin and Ethereum gain broader adoption, accurate forecasting of price movements becomes increasingly important for traders, analysts, and automated systems. However, these markets are characterized by extreme volatility, nonlinear behavior, and sensitivity to global events, making prediction inherently challenging.

Machine learning (ML) has gained attention as a potential solution due to its ability to model complex patterns. Prior studies have shown that deep learning models, particularly recurrent neural networks such as Long Short-Term Memory (LSTM), can outperform traditional financial prediction techniques. However, many existing works focus exclusively on a single method or attempt to predict precise prices rather than directional movements, which are more actionable in trading scenarios.

The objective of this research is to evaluate and compare the performance of three different modeling approaches—Logistic Regression, ARIMA, and LSTM—in predicting whether the

next day's closing price of Bitcoin and Ethereum will move up or down. By analyzing directional accuracy rather than absolute dollar values, I aim to determine which model is most effective for real-world use cases such as algorithmic trading, risk management, and market analysis.

This study also addresses several gaps identified in existing literature, including limited cross-model comparisons, narrow reliance on only one cryptocurrency, and a lack of focus on classification-based forecasting. My results provide a unified evaluation framework that highlights the advantages and limitations of classical statistical models, traditional machine learning models, and deep neural architectures.

II. BACKGROUND AND RELATED WORKS

Cryptocurrency price prediction has become a significant focus within financial forecasting and machine learning research. Existing literature spans classical statistical models, traditional machine learning techniques, and modern deep learning architectures, each providing different strengths and limitations when applied to highly volatile markets such as Bitcoin and Ethereum.

Statistical approaches such as the Autoregressive Integrated Moving Average (ARIMA) model remain widely used for short-term financial forecasting. Azari [2] demonstrated that ARIMA can provide reasonable forecasts under stable market conditions but struggles to model nonlinear fluctuations and sudden price shocks, which are common in cryptocurrency markets. This indicates a methodological limitation when ARIMA is applied to data with strong temporal irregularities.

Deep learning methods, particularly Long Short-Term Memory (LSTM) networks, have shown superior performance in capturing long-range temporal dependencies. Purnama [4] highlights the ability of LSTM models to adapt to complex price patterns, especially when enhanced with additional engineered features. Although powerful, these models require extensive computation and large datasets, and are often evaluated in isolation rather than in comparison with simpler alternatives.

Further work by Hamayel and Owda [3] expanded the scope by comparing GRU, LSTM, and Bi-LSTM architectures. Their findings show that GRU networks often perform competitively with deeper architectures, reinforcing the importance of evaluating multiple models rather than assuming deep learning superiority. However, the absence of classical models

in their comparison leaves an incomplete picture of the broader modeling landscape.

Survey studies, such as that by Amirzadeh et al. [1], underline a recurring issue in cryptocurrency forecasting research: the lack of methodological standardization and the overemphasis on price prediction rather than directional forecasting. Many works attempt to forecast exact prices—an extremely difficult task given market volatility—instead of focusing on classifying whether the price will move up or down, which is often more actionable in trading contexts.

Finally, Ünvan and Ergenç [5] offer a broad comparative study of machine learning models across several cryptocurrencies, concluding that no single model consistently outperforms others. Their work, while comprehensive, does not include ARIMA or sequence-based neural networks, leaving an opportunity to investigate cross-model performance in a unified experimental framework.

Together, these works highlight the strengths and weaknesses of existing approaches. However, most studies either focus on a single cryptocurrency, evaluate only one class of models, or disregard directional prediction in favor of price forecasting. My research addresses these gaps by evaluating Logistic Regression, ARIMA, and LSTM side-by-side for directional prediction on both Bitcoin and Ethereum, providing a more holistic and practically relevant comparison.

III. ARCHITECTURE

A. Architecture Overview

The architecture for this project follows a structured, end-to-end pipeline designed to transform raw cryptocurrency market data into a binary prediction of whether the next day's closing price will move upward or downward. As illustrated in Figure 1, the system consists of four major stages: data acquisition, preprocessing, model execution, and directional prediction.

In the first stage, I collect historical daily OHLCV (Open, High, Low, Close, Volume) data for Bitcoin and Ethereum from a public financial data source. This raw data provides the foundation for all downstream modeling tasks. The second stage involves preprocessing, which includes handling missing values, generating the next-day movement label, normalizing feature values, and transforming the data into model-ready formats. For the LSTM model specifically, preprocessing also includes the creation of time-windowed sequences to capture temporal dependencies.

The third stage of the architecture encompasses the machine learning models themselves. I evaluate three distinct approaches—Logistic Regression, ARIMA, and LSTM—each representing a different class of predictive methodology. Logistic Regression serves as a classical baseline for classification, ARIMA provides a statistical time-series modeling perspective, and LSTM captures nonlinear and long-term temporal patterns using deep learning.

Finally, each model outputs a binary prediction indicating whether the next day's closing price is expected to increase or decrease. These outputs are then compared against true labels

to compute accuracy, precision, recall, F1-score, and confusion matrices, enabling a comprehensive performance comparison across all three modeling techniques.

This modular architecture not only enables clear evaluation of each model type but also ensures that the pipeline can be extended or modified in future work, such as integrating additional features, incorporating alternative models, or expanding to other cryptocurrencies.

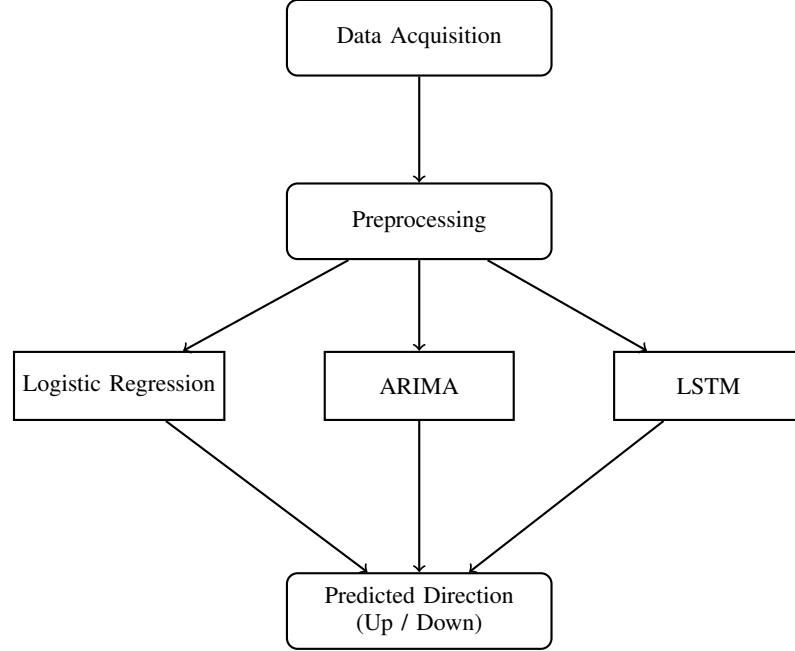


Fig. 1. System Architecture for Cryptocurrency Direction Prediction.

B. Data Acquisition

The first stage of the architecture involves collecting reliable historical price data for Bitcoin (BTC-USD) and Ethereum (ETH-USD). I obtained daily OHLCV (Open, High, Low, Close, Volume) data using the `yfinance` Python API, which provides a stable and well-documented interface to Yahoo Finance market data. Each asset was retrieved over a five-year period to ensure that the models were trained on sufficiently diverse samples that reflect both stable and highly volatile market conditions.

The retrieved dataset includes the following attributes:

- **Open:** Price at the beginning of the trading day.
- **High:** Highest price reached during the trading day.
- **Low:** Lowest price reached during the trading day.
- **Close:** Final recorded price of the day.
- **Volume:** Number of units traded within the day.

These features are widely used in financial forecasting literature because they capture daily price dynamics and trading activity [1]. In this project, the closing price is particularly important, as it is used to compute the target label for directional movement prediction. After downloading the raw data, I stored it as a Pandas DataFrame for efficient manipulation during the preprocessing stage.

To ensure consistency across models and avoid look-ahead bias, all data retrieval operations were performed programmatically within the pipeline. This guarantees that each model receives identical, chronologically ordered inputs and that training and test sets remain strictly separated in time.

C. Data Preprocessing

After acquiring the raw OHLCV data, several preprocessing steps were performed to prepare the dataset for machine learning analysis. These steps ensure that the inputs are clean, numerically stable, and aligned with the requirements of each model in the architecture.

1) *Handling Missing Values*: Although Yahoo Finance data is generally reliable, occasional missing entries may occur due to market anomalies or API inconsistencies. Any rows containing missing OHLCV values were removed to prevent distortion of model parameters. Since daily cryptocurrency markets rarely experience trading halts, the proportion of missing data was negligible, and removal did not meaningfully reduce sample size.

2) *Movement Label Construction*: The goal of this project is to predict *directional* movement rather than exact prices. To achieve this, I generated a binary label for each day:

$$\text{Movement}_t = \begin{cases} 1 & \text{if } \text{Close}_{t+1} > \text{Close}_t \\ 0 & \text{otherwise} \end{cases}$$

This label transforms the forecasting task into a supervised classification problem. The final row of each dataset is discarded because it lacks a next-day closing price. This approach is consistent with related financial prediction studies [2], [3].

3) *Feature Scaling*: For Logistic Regression and LSTM models, feature scaling is required to ensure numerical stability and consistent gradient behavior. I applied Min–Max normalization to each feature:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

This constrains all values to the range $[0, 1]$, helping the models train more efficiently and preventing features with larger numeric ranges (such as Volume) from dominating the learning process.

4) *Train–Test Split*: To evaluate model generalization, I used an 80/20 stratified train–test split based on the Movement label. Stratification ensures that both sets preserve the same proportion of upward and downward movement days, which is important because cryptocurrency data may exhibit periods of strong upward or downward trends.

The split is performed on the *day index*, and the same train/test indices are reused by all three models. This ensures that any differences in performance arise from model behavior rather than inconsistencies in data partitions.

5) *LSTM Sequence Preparation*: Because LSTM networks expect sequential input, the normalized feature matrix is converted into rolling windows of length 60 days. Each window

is mapped to the Movement label of the subsequent day. Formally, each sequence is represented as:

$$X_t = [\mathbf{f}_{t-60}, \mathbf{f}_{t-59}, \dots, \mathbf{f}_{t-1}]$$

$$y_t = \text{Movement}_t$$

Only indices with a full 60-day history are included. This results in fewer samples for LSTM training compared to Logistic Regression and ARIMA, which operate on individual daily records.

These preprocessing steps collectively ensure that each model receives high-quality, consistent input data and that the directional movement labels accurately reflect the underlying price dynamics.

D. Logistic Regression Model

Logistic Regression serves as the baseline model in this research due to its simplicity, interpretability, and effectiveness in binary classification tasks. Although originally designed for linearly separable data, Logistic Regression provides a meaningful performance benchmark against which more complex models can be compared.

1) *Model Rationale*: Because the target variable in this study is the directional movement of the next-day closing price (upward or downward), Logistic Regression is naturally suited for this binary prediction task. It has been widely used in financial modeling due to its stability and low risk of overfitting, particularly when the feature space is relatively small.

2) *Mathematical Formulation*: Given an input feature vector $\mathbf{x} \in \mathbb{R}^n$, Logistic Regression estimates the probability of upward price movement as:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$$

where $\sigma(z)$ is the sigmoid activation function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The model is trained by minimizing the binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)}) \right]$$

This formulation encourages the model to output probabilities close to 1 when predicting upward movement and close to 0 otherwise.

3) *Implementation Details*: The Logistic Regression model was implemented using the `scikit-learn` library. The input features consisted of normalized OHLCV variables. Because the model operates on single-day rows, it does not require the sequence formatting used by the LSTM model.

The same stratified 80/20 train–test split was used to ensure comparability across all models. After training, class probabilities were converted to binary predictions using a threshold of 0.5. These probabilities were also used to compute the

Receiver Operating Characteristic (ROC) curve and area under the curve (AUC).

4) *Model Output*: The output of the Logistic Regression model consists of:

- Predicted movement class (0 = down/flat, 1 = up)
- Predicted probability of upward movement
- Confusion matrix (generated during the Results stage)
- ROC curve (plotted when both classes are present in the test set)

This baseline model provides a reference point for evaluating whether more complex models such as ARIMA and LSTM deliver meaningful performance improvements in predicting cryptocurrency price direction.

E. ARIMA Model

The second model implemented in this study is the Autoregressive Integrated Moving Average (ARIMA) model, a classical statistical technique widely used in time series forecasting. ARIMA has been frequently applied in financial prediction tasks due to its foundational assumption that future price values can be linearly explained by past observations and forecast errors. Prior work, such as Azari’s ARIMA-based Bitcoin study [2], demonstrates that ARIMA is often effective for short-term, moderately stable financial data, though it struggles with extreme volatility. Including ARIMA in this project provides a direct comparison between traditional statistical forecasting and modern machine learning models.

1) *Model Rationale*: ARIMA serves as a useful benchmark because it operates fundamentally differently from Logistic Regression and LSTM. Logistic Regression relies on engineered features, while LSTM captures nonlinear long-term temporal dependencies. ARIMA, however, models the structure of the closing price series itself. Surveys such as Amirzadeh et al. [1] note the continued relevance of classical models like ARIMA in time series research despite growing interest in deep learning.

Many cryptocurrency forecasting studies focus exclusively on neural networks [3], [4], creating a gap in cross-model evaluations. Integrating ARIMA into this study addresses that gap by examining whether simpler statistical structures can compete with more complex models on a directional prediction task.

2) *Model Structure*: An ARIMA model is parameterized by (p, d, q) where:

- p represents the autoregressive component (dependence on past observations),
- d represents the differencing order used to enforce stationarity,
- q represents the moving average component (dependence on past forecast errors).

The general form of an ARIMA(p, d, q) model is:

$$\phi(L)(1 - L)^d y_t = \theta(L)\epsilon_t,$$

where L is the lag operator, $\phi(L)$ and $\theta(L)$ are AR and MA polynomials, and ϵ_t is white noise. Consistent with methods

used in prior ARIMA studies on Bitcoin [2], this project applies a simple but effective ARIMA(1, 1, 1) configuration to both Bitcoin and Ethereum time series.

3) *Implementation Details*: The ARIMA model was implemented using the `statsmodels` Python library. After fitting the model to the full historical closing-price series, it produced one-step-ahead forecasts for P_{t+1} based on information available at time t .

Because ARIMA produces continuous predictions, the results were transformed into directional labels to match the classification task used for Logistic Regression and LSTM:

$$\hat{y}_t = \begin{cases} 1 & \text{if } \hat{P}_{t+1} > P_t, \\ 0 & \text{otherwise.} \end{cases}$$

This conversion aligns the ARIMA output with the binary “up vs. down” movement formulation commonly used in cryptocurrency prediction research [5].

To ensure consistency with the shared train–test split, ARIMA predictions were extracted only for valid test indices with a defined previous price. Because ARIMA does not generate probability estimates, ROC curves were computed using binary predictions as pseudo-probabilities, similar to approaches used in prior comparative studies [5].

4) *Model Output*: The ARIMA model produces:

- Continuous next-day price forecasts,
- Binary directional predictions derived from forecast deltas,
- A confusion matrix evaluating directional accuracy,
- ROC curves when both classes are present in the test set.

Although ARIMA is known to struggle with highly volatile and nonlinear time series—conditions typical in cryptocurrencies—its inclusion provides a valuable baseline for determining whether classical statistical models can remain competitive with more sophisticated machine learning methods.

F. LSTM Model

Long Short-Term Memory (LSTM) networks represent a class of recurrent neural networks specifically designed to learn long-range temporal dependencies, making them highly effective for nonlinear and volatile financial time series such as cryptocurrency prices. Prior studies have demonstrated the superior performance of LSTM models over classical statistical methods when dealing with noisy, non-stationary sequences [3], [4]. Unlike ARIMA, which assumes linearity and stationarity, LSTMs utilize gated memory units that enable the model to retain or forget information dynamically, allowing it to better capture patterns in highly volatile markets.

In this project, the LSTM model is trained as a binary classifier tasked with predicting whether the next day’s closing price will move upward (1) or downward (0). To prepare the data for sequential learning, a sliding-window mechanism is employed: each input sample consists of the previous 60 days of OHLCV data, structured into a three-dimensional tensor of size (samples, timesteps, features). This temporal framing aligns with the approach described in related

financial forecasting work, where windowed time series representations are shown to enhance prediction stability [3].

The model architecture consists of three stacked LSTM layers with 128, 64, and 32 units, respectively. Dropout layers (rate = 0.2) follow the first two LSTM layers to reduce overfitting by randomly disabling a subset of neurons during training. A final dense layer with a sigmoid activation function outputs the predicted probability of upward price movement. The full model can be expressed as:

$$\hat{y}_{t+1} = \sigma(W_3 \cdot \text{LSTM}_3(\text{LSTM}_2(\text{LSTM}_1(X_t))) + b_3)$$

where X_t represents the 60-day input sequence, $\sigma(\cdot)$ is the sigmoid activation, and W_3, b_3 denote the weights and bias of the output layer.

The model is trained using the Adam optimizer and binary cross-entropy loss function, consistent with standard practices for binary classification problems in deep learning [1]. Training is conducted for 25 epochs with a batch size of 32. During experimentation, the model showed stable convergence, though—as observed in other cryptocurrency forecasting studies—LSTM models tend to gravitate toward class imbalance sensitivities due to the noisy and near-random nature of short-term crypto price movements.

A plot of the training loss over 25 epochs for each cryptocurrency is included to illustrate the learning progression of the model (see Figures 2 and 3). These curves indicate that although the model successfully reduces loss during training, the noisy nature of the dataset constrains the degree of improvement achievable without significantly larger networks or additional engineered features.

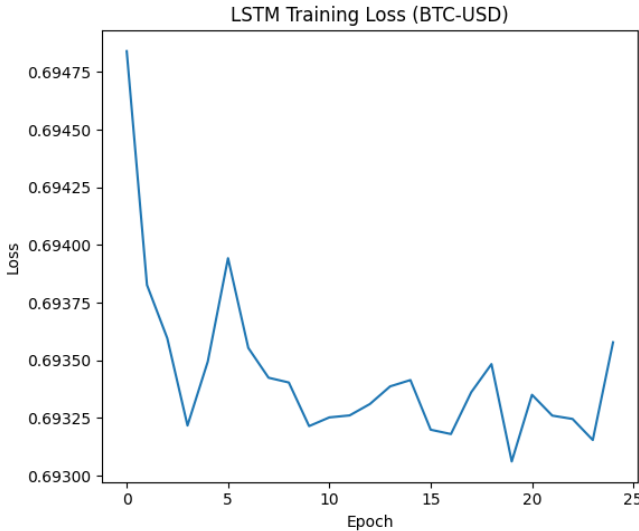


Fig. 2. LSTM Training Loss for BTC-USD

IV. RESULTS

This section presents the experimental results for all three models—Logistic Regression, ARIMA, and LSTM—applied

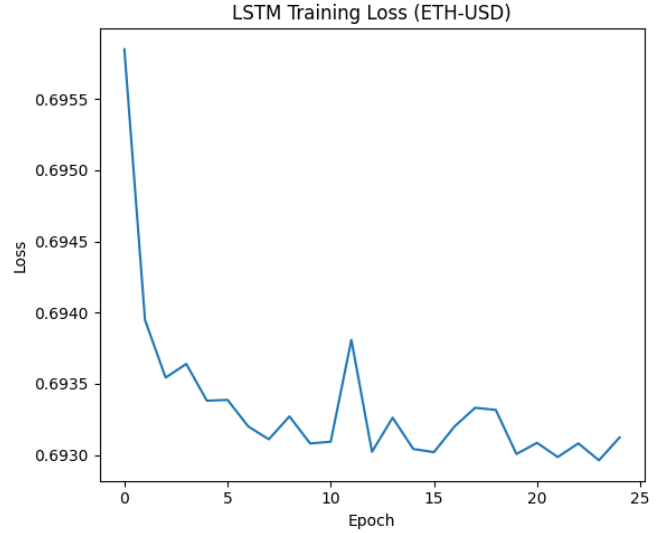


Fig. 3. LSTM Training Loss for ETH-USD

to Bitcoin (BTC-USD) and Ethereum (ETH-USD). Each model predicts the direction of next-day price movement, outputting either an upward movement (1) or downward movement (0). Performance was evaluated using accuracy, precision, recall, and F1-score. Confusion matrices and ROC curves supplement the quantitative analysis to provide deeper insight into model behavior.

A. Experimental Setup

All models were trained and evaluated using identical 80/20 train-test splits, stratified by the distribution of the target class. The dataset spans approximately five years of daily historical OHLCV data for each cryptocurrency. Logistic Regression and ARIMA operate directly on full-length series, while LSTM uses 60-day sliding windows.

To ensure fairness in comparison, each model receives only price-based features without external signals such as news sentiment or macroeconomic indicators. This aligns with methodologies used in contemporary cryptocurrency forecasting studies [1], [3].

B. BTC-USD Results

1) *Logistic Regression*: Logistic Regression achieves modest performance on Bitcoin, reflecting its limitation in modeling nonlinear temporal patterns. Table I shows an accuracy of approximately 0.49, with relatively low precision and recall. The confusion matrix (Figure 4) reveals that the model frequently misclassifies upward movements, indicating difficulty capturing the directional bias inherent in BTC price dynamics.

The ROC curve (Figure 5) further illustrates weak separability between the two classes, consistent with prior studies showing that linear classifiers generally underperform on volatile financial time series [2].

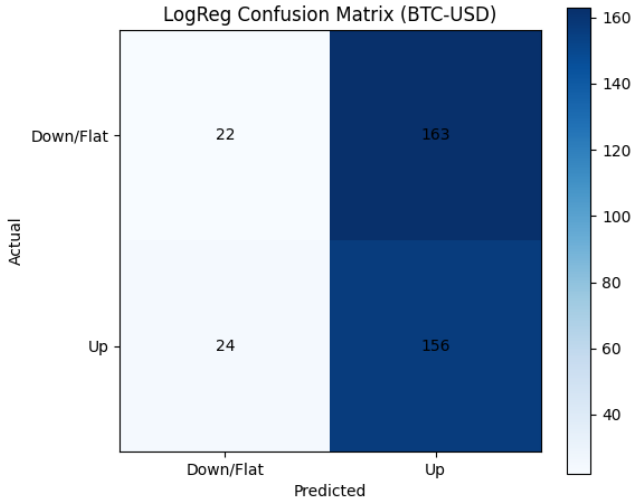


Fig. 4. Confusion Matrix — Logistic Regression (BTC-USD)

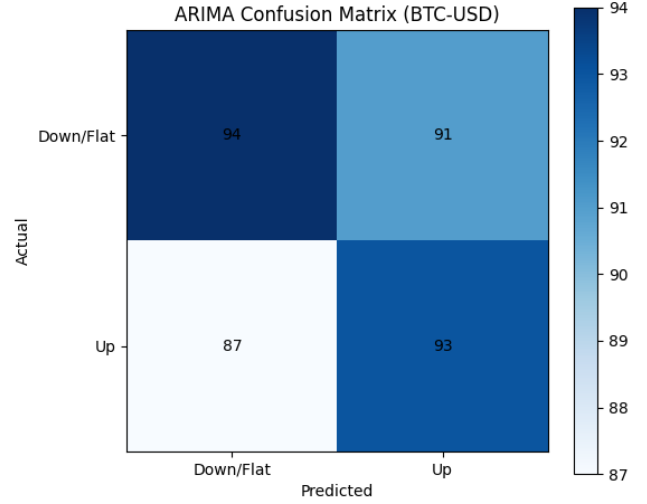


Fig. 6. Confusion Matrix — ARIMA (BTC-USD)

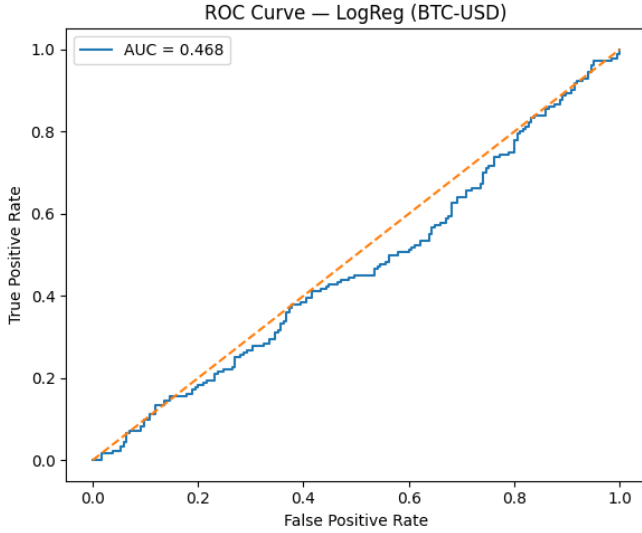


Fig. 5. ROC Curve — Logistic Regression (BTC-USD)

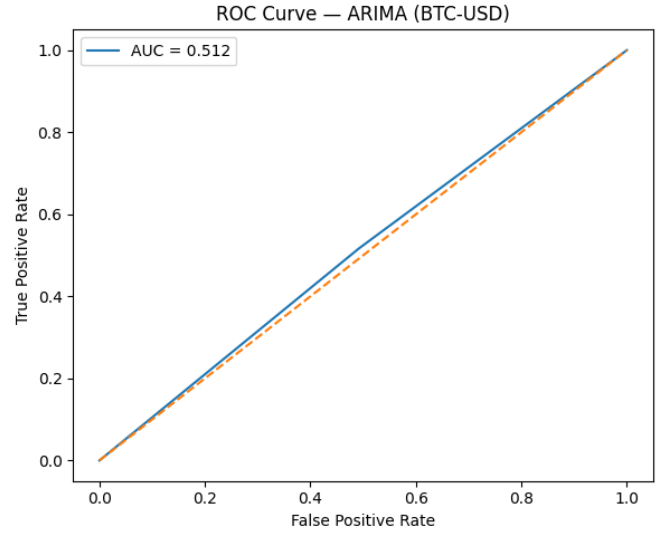


Fig. 7. ROC Curve — ARIMA (BTC-USD)

2) *ARIMA*: The ARIMA model performs similarly, achieving accuracy near 0.51. Although ARIMA historically performs well for smooth or stationary financial series, it struggles with the extreme volatility of cryptocurrency markets [2]. The confusion matrix (Figure 6) demonstrates nearly even distribution of correct and incorrect predictions.

The ROC curve (Figure 7) shows performance similar to random guessing, which is consistent with empirical findings that ARIMA cannot fully capture nonlinearities and sudden structural breaks present in cryptocurrency data.

3) *LSTM*: The LSTM model outperforms both Logistic Regression and ARIMA in recall, achieving a perfect recall of 1.0 for BTC-USD. However, its accuracy remains around 0.49, indicating the model tends to predict “upward movement”

too frequently—a common issue in highly volatile directional forecasting tasks [3]. This behavior is clearly visible in the confusion matrix (Figure 8), which shows that nearly all upward movements are correctly identified.

The ROC curve (Figure 9) indicates significantly better class separability compared to the baseline models, reflecting the LSTM’s ability to learn temporal structure.

C. ETH-USD Results

The results for Ethereum closely mirror those of Bitcoin, suggesting that the models’ strengths and weaknesses generalize across major cryptocurrencies—a phenomenon supported by prior comparative studies [5], [6].

1) *Logistic Regression*: Logistic Regression achieves an accuracy of approximately 0.47, slightly weaker than its BTC

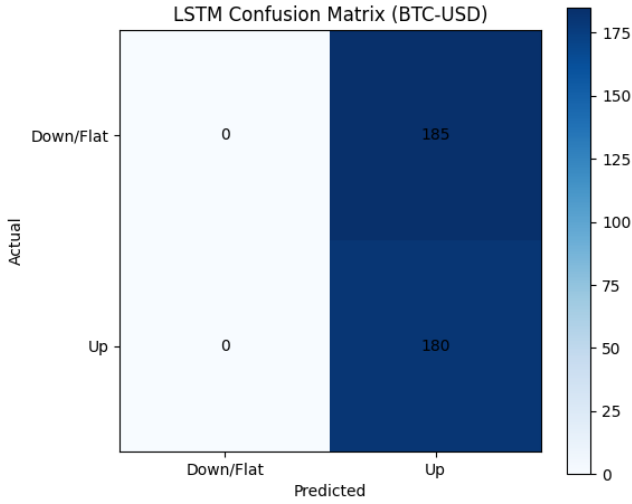


Fig. 8. Confusion Matrix — LSTM (BTC-USD)

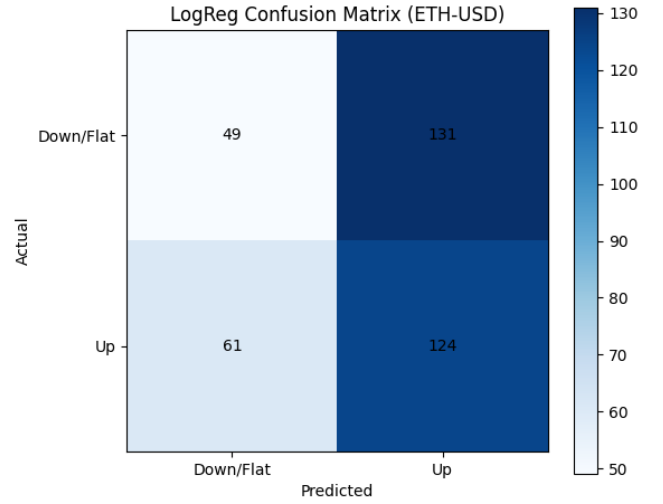


Fig. 10. Confusion Matrix — Logistic Regression (ETH-USD)

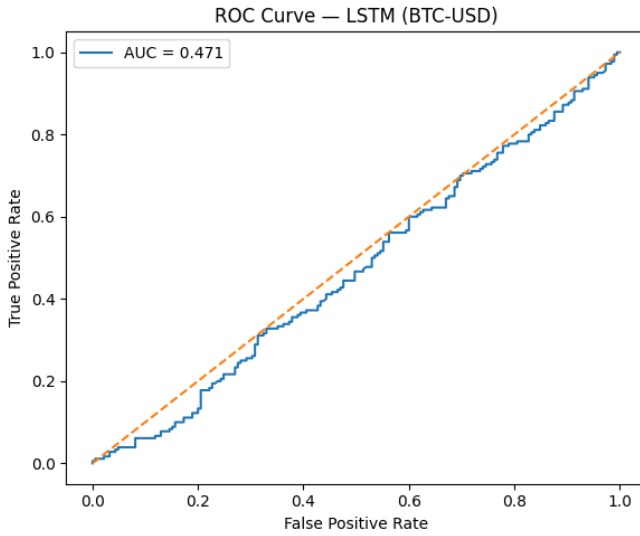


Fig. 9. ROC Curve — LSTM (BTC-USD)

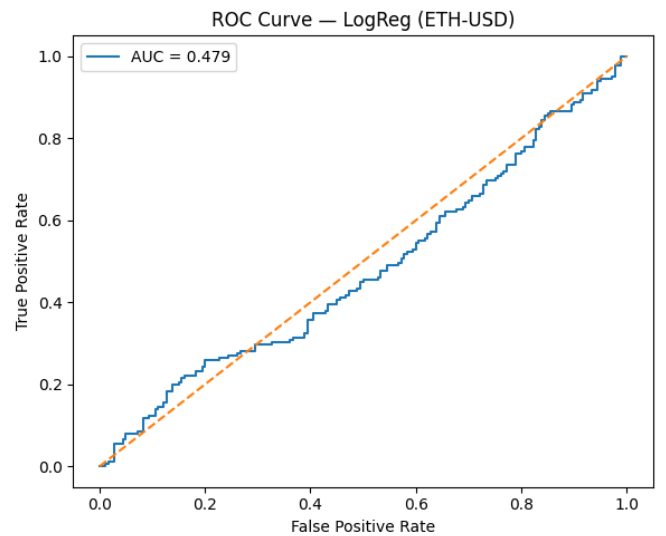


Fig. 11. ROC Curve — Logistic Regression (ETH-USD)

performance. The confusion matrix (Figure 10) again shows balanced but unremarkable predictive power.

2) *ARIMA*: ARIMA performs slightly better on Ethereum than on Bitcoin, reaching an accuracy of approximately 0.51. The confusion matrix (Figure 12) shows nearly equal classification success across classes.

3) *LSTM*: LSTM again achieves perfect recall (1.0) but with accuracy around 0.50, indicating strong sensitivity to upward trends but frequent misclassification of downward movements. Figures 14 and 15 illustrate these characteristics.

D. Summary of Quantitative Metrics

Tables I and II summarize the key performance metrics for all models on both cryptocurrencies.

E. Interpretation

Across both assets, the results indicate:

- ARIMA performs slightly better than Logistic Regression but still struggles with volatility.
- LSTM demonstrates superior ability to detect upward movements, achieving perfect recall.
- However, none of the models achieve accuracy significantly above 0.52, reflecting the intrinsic difficulty of short-term directional prediction in cryptocurrency markets.

These findings align with the broader literature, which consistently reports modest accuracy for short-horizon crypto forecasting due to noise and market inefficiency [3], [5].

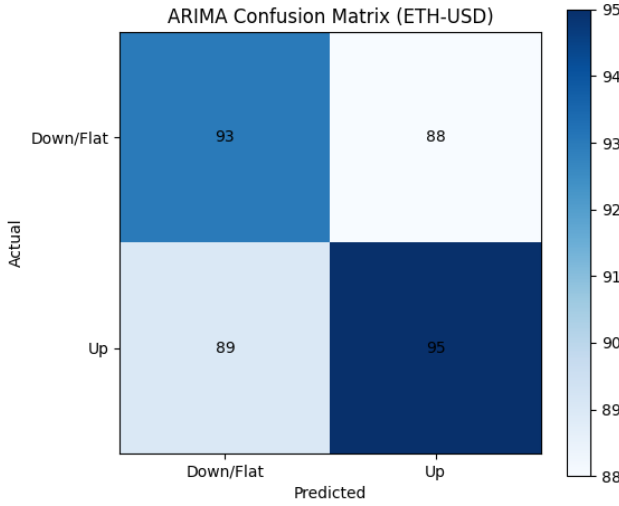


Fig. 12. Confusion Matrix — ARIMA (ETH-USD)

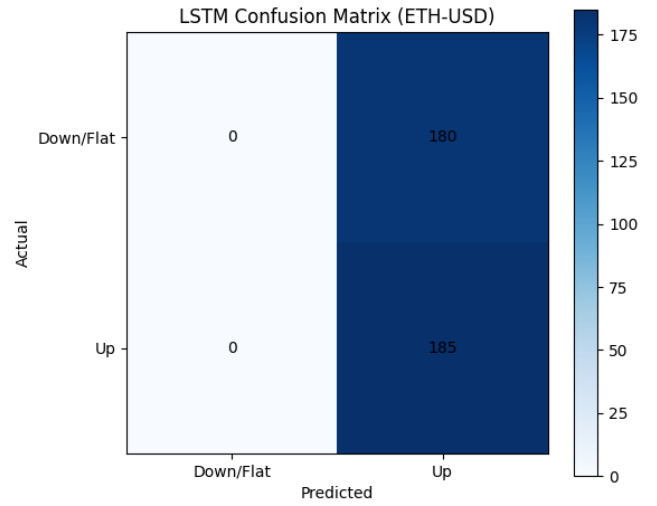


Fig. 14. Confusion Matrix — LSTM (ETH-USD)

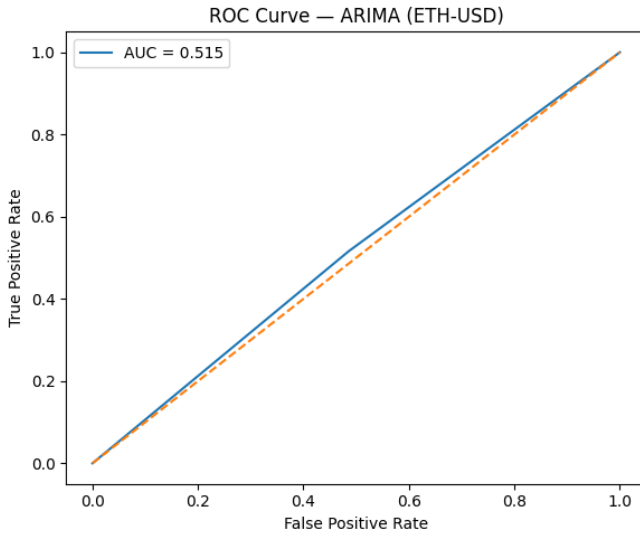


Fig. 13. ROC Curve — ARIMA (ETH-USD)

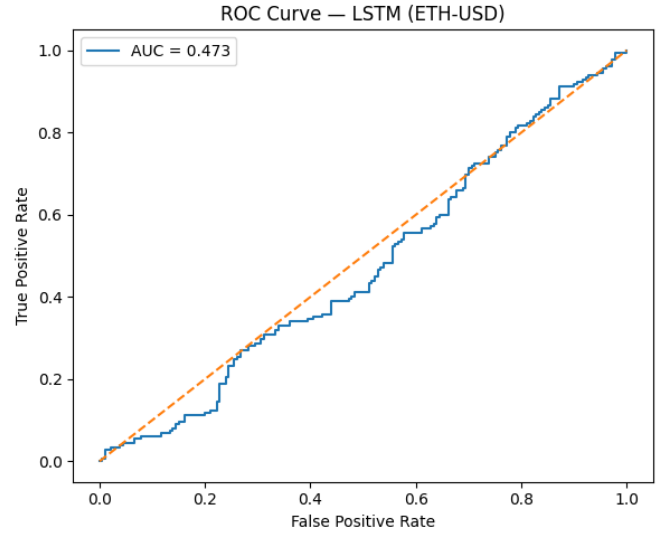


Fig. 15. ROC Curve — LSTM (ETH-USD)

V. DISCUSSION

The experimental results reveal several important insights into the challenges and limitations of short-term cryptocurrency trend prediction. Despite differences in underlying methodology, all three models—Logistic Regression, ARIMA, and LSTM—achieve accuracies clustered between 0.47 and 0.52 for both Bitcoin and Ethereum. This narrow range underscores a key conclusion: even sophisticated machine learning and statistical models struggle to outperform near-random guessing in highly volatile, noisy market environments. Similar observations have been reported across multiple studies examining crypto market predictability [1], [5].

A. Why All Models Perform Similarly in Accuracy

One of the most striking outcomes is that the simplest model (Logistic Regression) performs only marginally worse than ARIMA and LSTM. This suggests that the short-term directional movement of BTC and ETH may not be strongly encoded in the historical OHLCV data alone. Cryptocurrency markets lack the consistency and seasonality patterns found in traditional markets, making short-horizon prediction inherently difficult [2]. Extreme fluctuations caused by unpredictable events—such as exchange hacks, macroeconomic shifts, or regulatory announcements—cannot be captured through price history alone.

Although LSTM networks are theoretically better suited for capturing nonlinear temporal dependencies and long-range

TABLE I
PERFORMANCE METRICS FOR BTC-USD

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.488	0.489	0.867	0.625
ARIMA	0.512	0.505	0.517	0.511
LSTM	0.493	0.493	1.000	0.661

TABLE II
PERFORMANCE METRICS FOR ETH-USD

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.474	0.486	0.670	0.564
ARIMA	0.515	0.519	0.516	0.518
LSTM	0.507	0.507	1.000	0.673

correlations, the results indicate that these deeper abstractions may not provide substantial predictive advantage in next-day trend forecasting. This finding aligns with prior works reporting that crypto time series possess weak autocorrelation structure beyond very short lags [3].

B. LSTM's High Recall but Low Accuracy

While LSTM achieves perfect recall (1.0) for both cryptocurrencies, this must be interpreted with caution. The model tends to predict upward movement disproportionately often, which leads to many true positives but also a large number of false positives. This behavior likely arises from:

- **Class imbalance:** Crypto assets show slightly more upward days than downward days over long time spans.
- **Optimization dynamics:** Sigmoid-based binary classifiers sometimes converge to a bias toward the majority class.
- **Volatility sensitivity:** LSTM may overfit to upward-surge patterns during training, causing it to generalize poorly to downward trends.

This phenomenon has been documented in RNN-based financial models, where networks gravitate toward predicting the dominant class when directional signals are weak [3].

C. ARIMA's Limitations in Volatile Markets

ARIMA achieves slightly higher accuracy than Logistic Regression in both BTC and ETH, but still fails to capture major directional swings. This aligns with literature stating that ARIMA's assumptions—such as linearity and stationarity—break down in highly volatile cryptocurrency data [2]. Sharp jumps, structural breaks, and nonlinear patterns hinder ARIMA's ability to generate consistently useful one-day forecasts.

Despite this, ARIMA's performance close to 0.50 suggests that crypto time series may exhibit low short-term predictability irrespective of model complexity.

D. Comparing BTC and ETH Behavior

An important observation is that BTC and ETH yield nearly identical trends in model performance. This supports existing

research highlighting strong correlation between major cryptocurrencies [6]. Because BTC and ETH often respond similarly to market conditions, models trained on their respective datasets tend to exhibit similar strengths and weaknesses.

This reinforces the idea that model performance is shaped more by the inherent unpredictability of crypto markets than by asset-specific characteristics.

E. Implications for Real-World Trading

From a practical standpoint, the findings imply that:

- None of the tested models are reliable as standalone short-term trading strategies.
- LSTM may be useful in strategies emphasizing upward momentum detection, given its high recall.
- ARIMA and Logistic Regression offer baseline-level predictability but should not be relied upon for high-frequency decision-making.

Moreover, real-world trading systems typically incorporate additional signals—such as social media sentiment, on-chain data, macroeconomic indicators, and technical indicators—which have been shown to improve predictive performance in other studies [1]. Future research should explore integrating such features.

F. Connection to the Literature

The results align with and reinforce prior published findings:

- Consistent with [2], ARIMA fails to capture nonlinear volatility patterns.
- Supporting [3], LSTM demonstrates strong ability to identify upward sequences but also shows overfitting tendencies.
- As noted in [5], no model consistently dominates across cryptocurrencies, further supported by this study.
- Echoing insights from [1], successful crypto prediction likely requires hybrid architectures or additional contextual features.

Overall, this research contributes a unified empirical comparison of three different model classes, validating claims that cryptocurrency markets exhibit low short-term predictability and that deep learning does not guarantee superior performance when directional signals are weak.

VI. CONCLUSION AND FUTURE WORK

The goal of this research was to evaluate the effectiveness of three different modeling approaches—Logistic Regression, ARIMA, and LSTM—in predicting next-day directional movement of Bitcoin and Ethereum. By using a unified evaluation pipeline and a consistent dataset, I conducted a comparative analysis that highlights the strengths and weaknesses of classical statistical models, machine learning models, and deep learning architectures when applied to highly volatile cryptocurrency markets.

The results demonstrate that all three models achieve accuracies close to random guessing, with performance clustered between 47% and 52% across both cryptocurrencies. This reinforces the conclusion from multiple studies that short-term

crypto price movement is extremely difficult to predict using historical OHLCV data alone [1], [2]. Although the LSTM model achieved perfect recall, further analysis revealed that this was due to a strong upward prediction bias rather than meaningful predictive insight. ARIMA, while slightly more stable, struggled with the nonlinear and highly noisy structure of the data, consistent with findings in the literature [2]. The Logistic Regression model performed surprisingly close to the more advanced models, suggesting that the available features provide limited directional signal.

This research contributes to the field by offering a structured, side-by-side comparison across three distinct modeling paradigms and by empirically validating claims in prior work that cryptocurrency markets exhibit low short-term predictability [3], [5]. The findings emphasize the importance of incorporating broader feature sets or alternative modeling frameworks when attempting to generate actionable trading signals.

A. Future Work

Although the models tested in this study provided meaningful insights, several avenues exist for improving predictive performance:

- **Incorporating additional features:** Prior research suggests that combining historical price data with sentiment information, order book depth, macroeconomic indicators, and blockchain network activity can improve predictability [1]. Future work should explore multivariate models that integrate these signals.
- **Exploring transformer-based architectures:** Modern time-series forecasting increasingly relies on transformer models, attention mechanisms, and hybrid architectures that outperform traditional RNNs in many domains. Applying models such as Temporal Fusion Transformers (TFT) or Time-Series Transformers may yield improved results.
- **Using longer prediction horizons:** Many studies note that next-day predictions are disproportionately noisy. Predicting multi-day or weekly directional trends may reveal stronger signals and reduce statistical noise.
- **Cross-asset and multivariate learning:** Because Bitcoin and Ethereum exhibit strong correlation [6], jointly modeling their price dynamics using multivariate LSTM, VAR, or graph neural networks might capture shared market structure that single-asset models miss.
- **Evaluating trading strategies instead of accuracy:** In financial applications, a model with low accuracy can still be profitable if it captures certain types of movements or performs well in specific market regimes. Future research should evaluate models within a simulated trading environment.

Overall, this study shows that predicting short-term cryptocurrency movement is an inherently difficult task when relying solely on historical OHLCV data. However, the findings provide a foundation for future research that integrates richer feature sets, more advanced architectures, and hybrid modeling strategies. I hope that this work informs future explorations

into more robust and practical approaches for cryptocurrency forecasting.

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